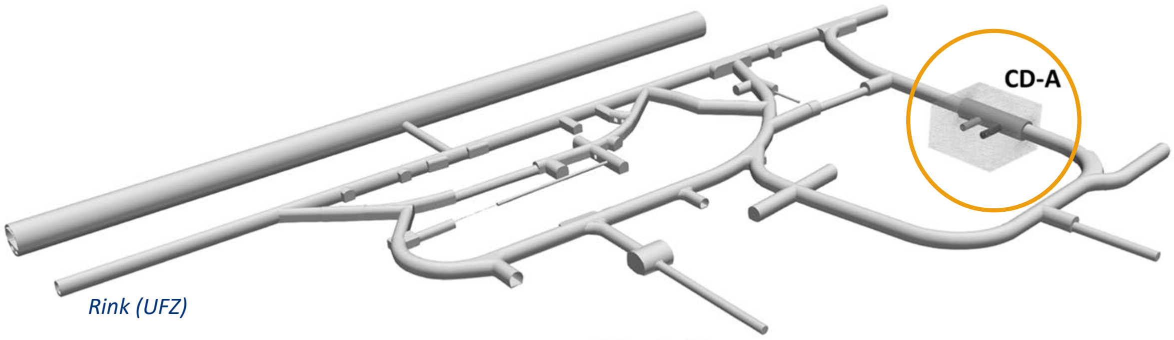
The CD-A experiment in Mont Terri

Mont Terri Project

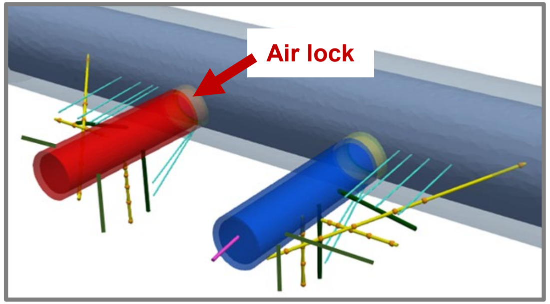
BGR

<u>Outline:</u>

- **Set-up of CD-A Experiment**
- **Measurements**
  - Long-term monitoring of electrical resistivity (ERT)
  - Measurements of water content (NMR)
- **Modelling CD-A**



Rink (UFZ)



CD-A Experiment:
Investigation of coupled HM effects in Opalinus clay

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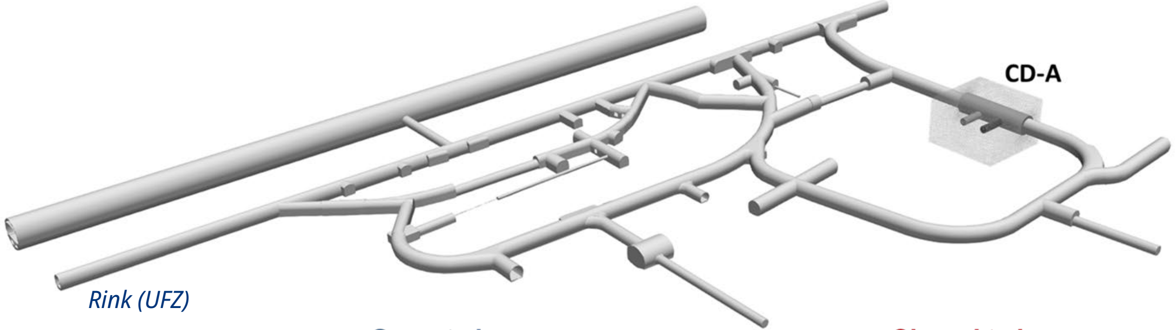
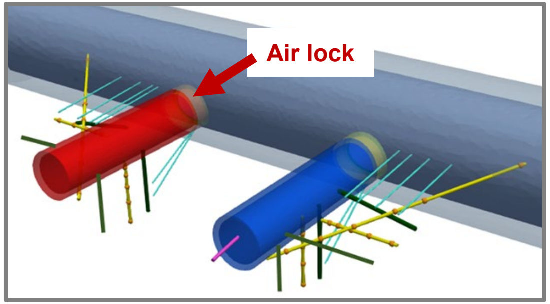


Rink (UFZ)



- 2 horizontal excavations
- perpendicular to strike of bedding
- no stabilization
- upper sandy facies
- influence of climatic conditions on HM behaviour
- Long-term investigation
- CD: “cyclic deformations”

Open twin:

Closed twin:

Open twin

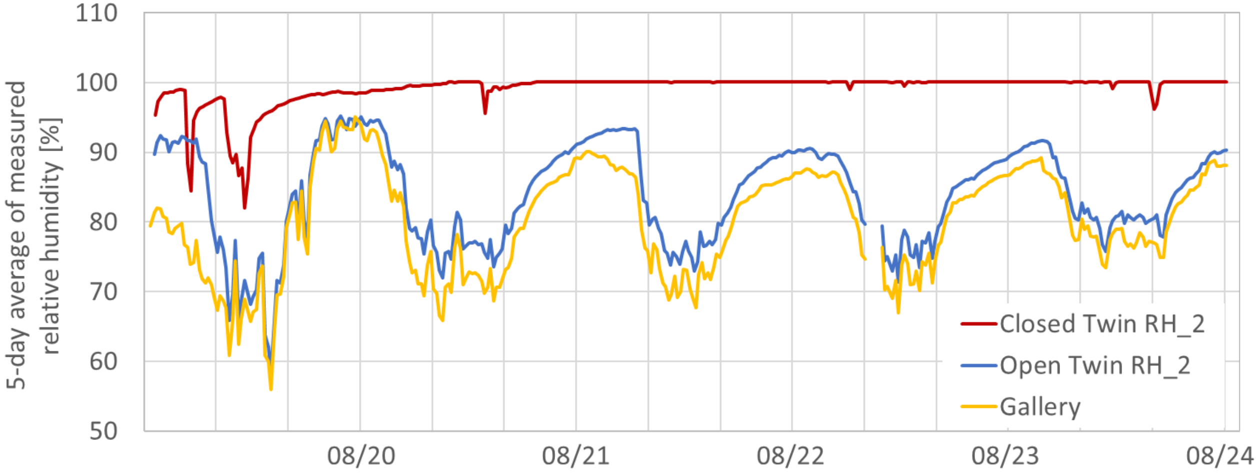
„natural“ niche
containing coupled HM effects due to desaturation

Closed twin

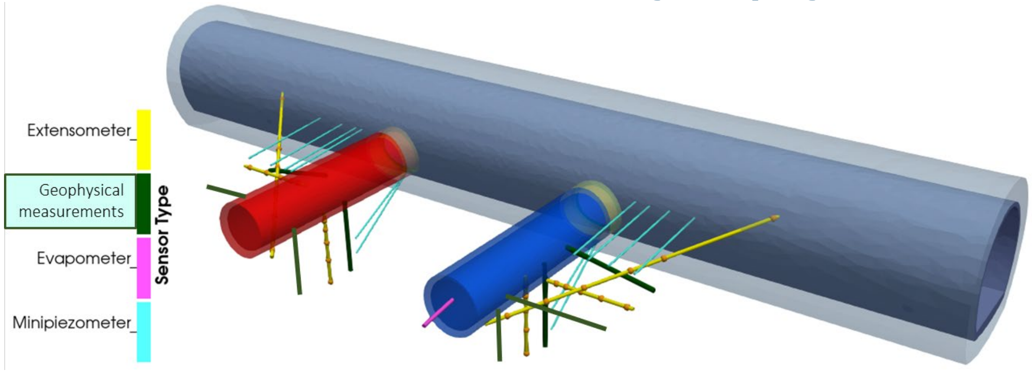
„air conditioned“ niche
avoiding desaturation effects as much as possible

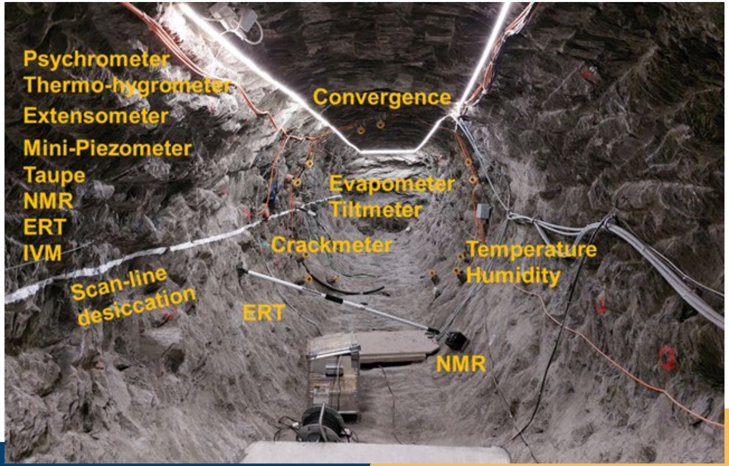
Aiming on a detailed **process** and **system understanding** as a basis for safety assessment.

Climatic conditions within the niches



Measuring campaign

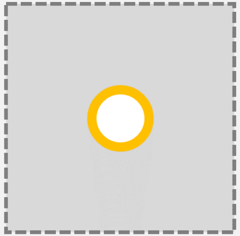




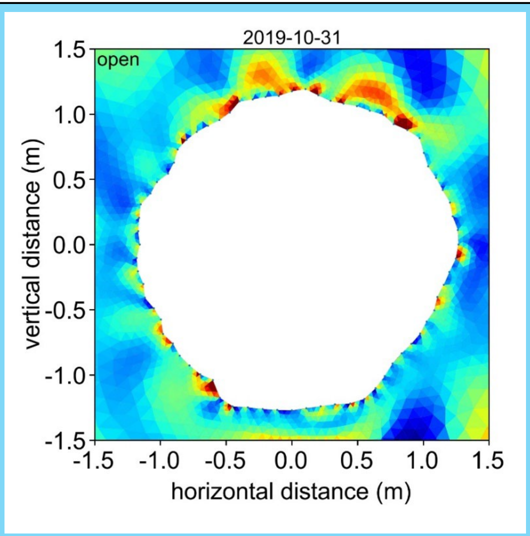
New results of ERT monitoring

(Markus Furche, BGR)

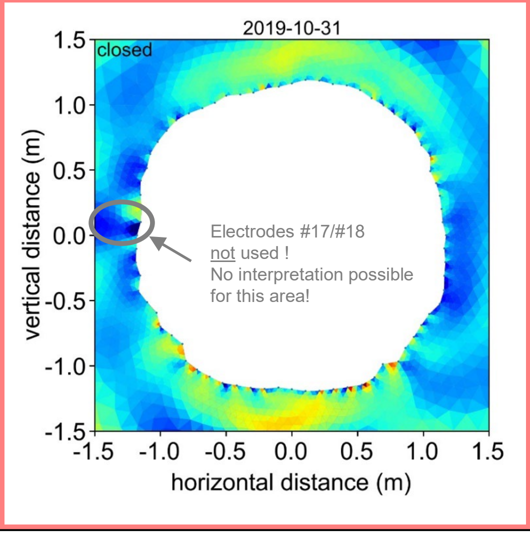
Topic:	Characterization of claystone
Measured value:	Electric resistivity
Unit:	Ωm
Method:	Electric resistivity tomography (ERT)
Interval:	Daily
Info:	72 equidistant electrodes along niche profile

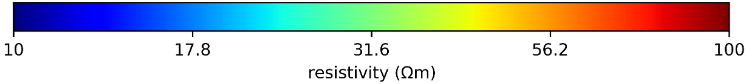


Niche open



Niche closed

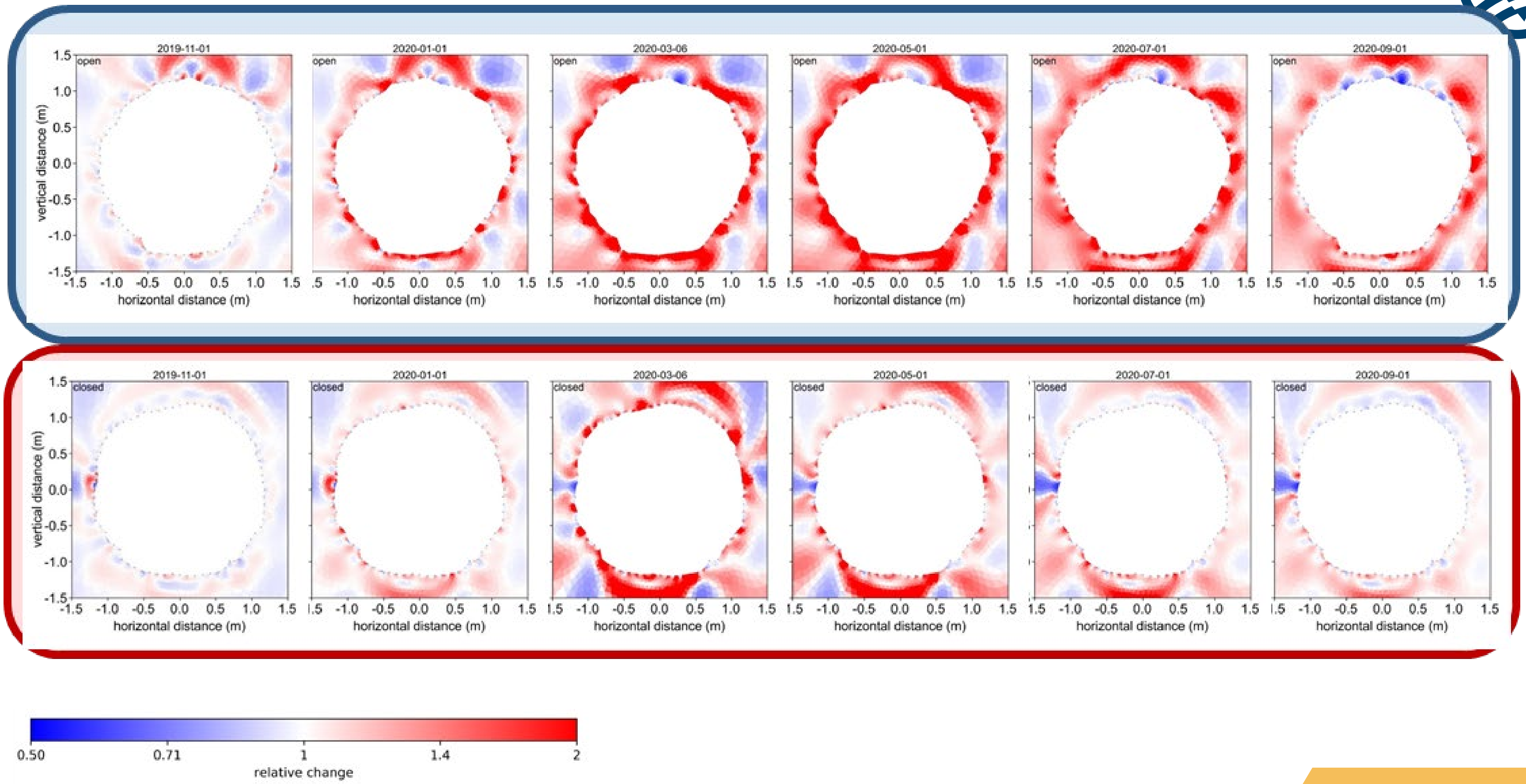




Reference day: 31.10.2019
(5 days after installation)

Furche (BGR)

Relative change of the electrical resistivity



Water content evolution at a depth of 40cm

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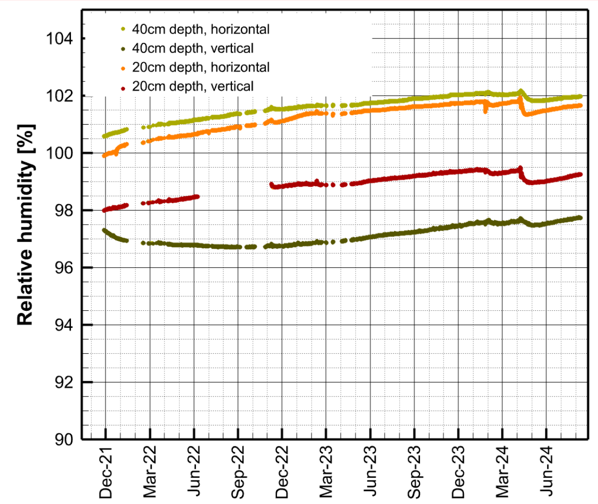

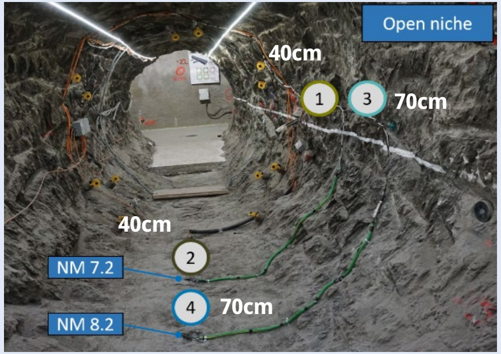

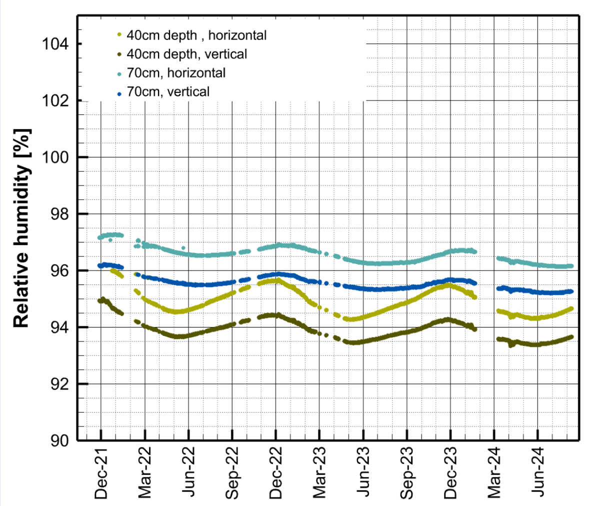

Closed niche:
- Higher humidity
- Increase of humidity

Open niche:
- Lower humidity
- Seasonality
- Decrease of humidity

Both niches:
Higher relative humidities in horizontal direction (lighter colors)

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Material parameters: Permeability measurements by pulse tests

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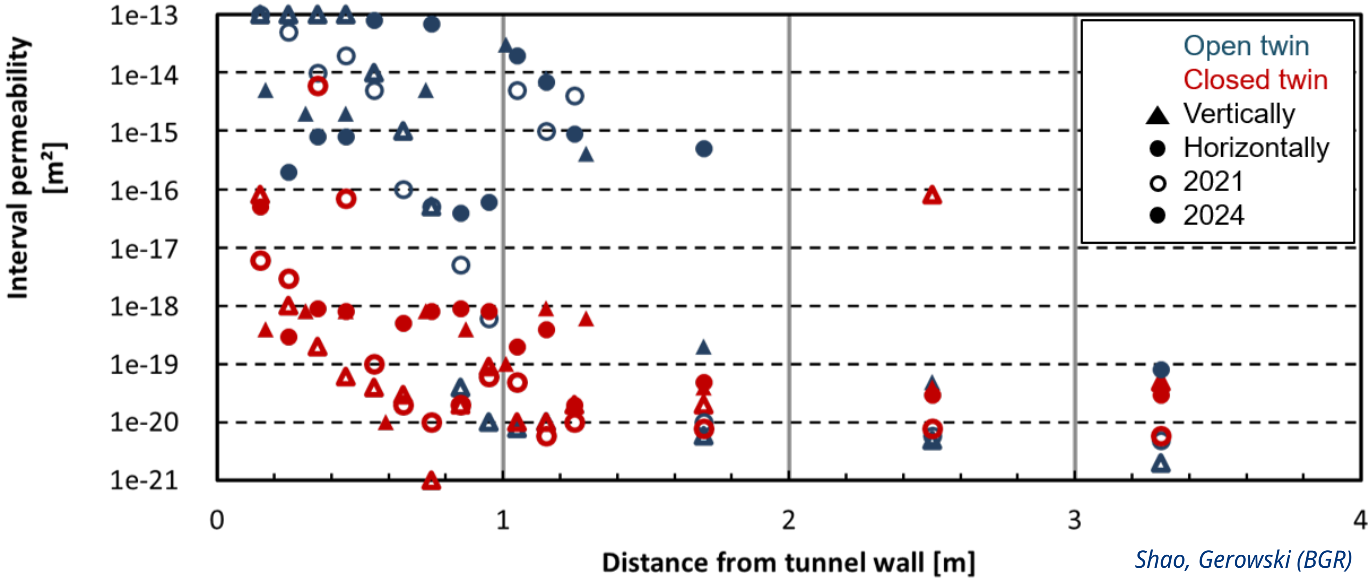
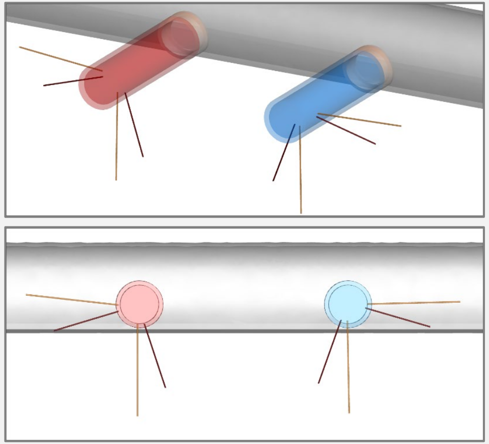


Shao, Gerowski (BGR)



Measurements in 2021
(1.5 yrs after excavation)
Measurements in 2024
(4.5 yrs after excavation)

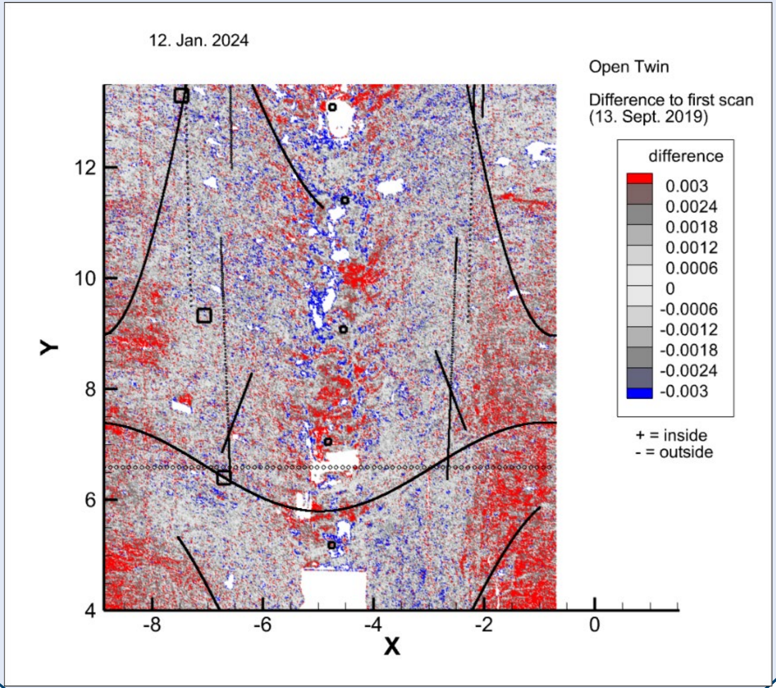
Significant differences up to 1.5 meters
Open twin: Permeability ~ 1e-15m² *Closed twin: Permeability ~ 1e-19m²*

Difference even clearer in vertical direction

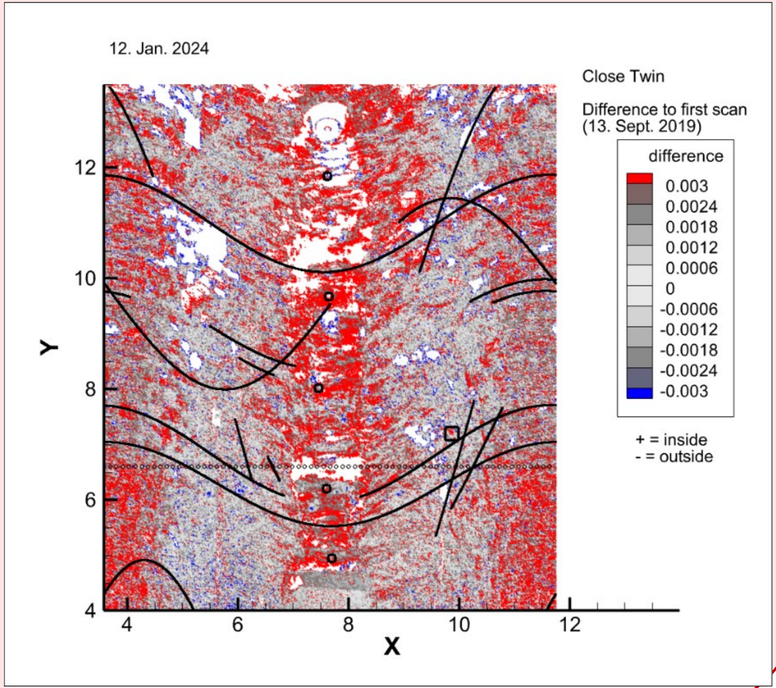
Difference becomes more evident over time

Convergence behavior: Laser scans of the niche surface

Open twin



Closed twin



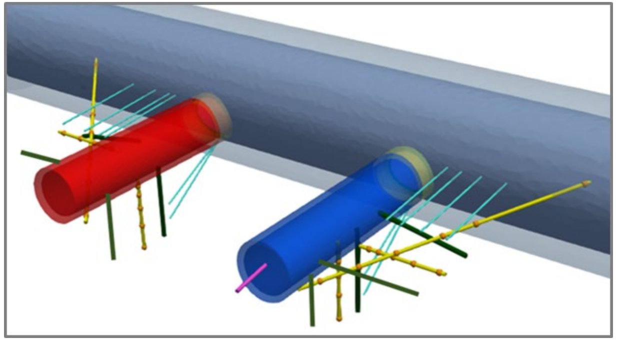
Fault zones

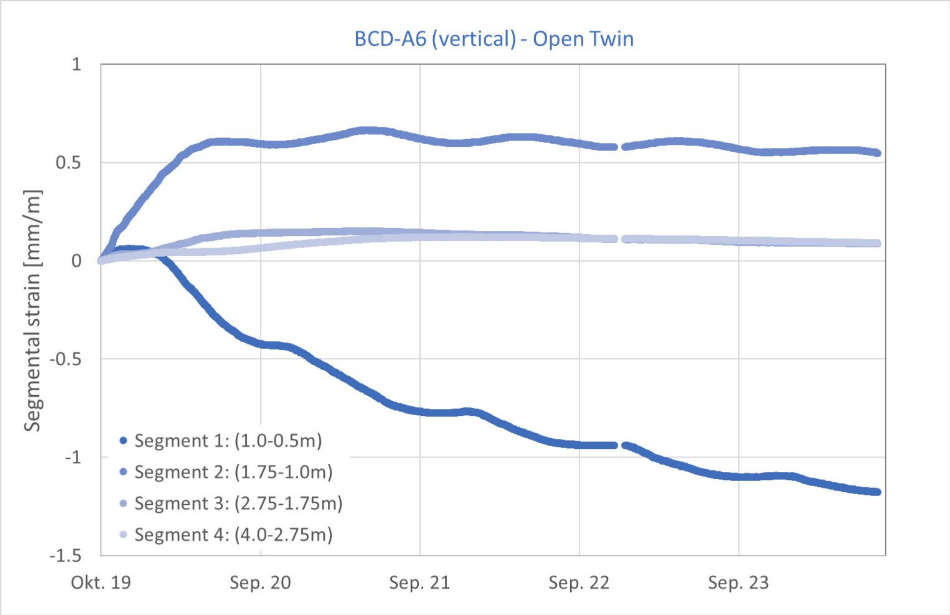
Convergence > 3mm

Divergence > 3mm

Convergence behavior is more pronounced in closed than in open twin

Segmental strains under the niches

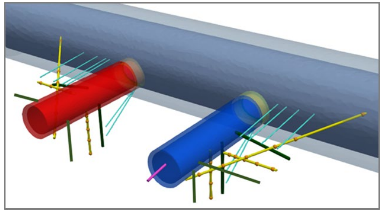


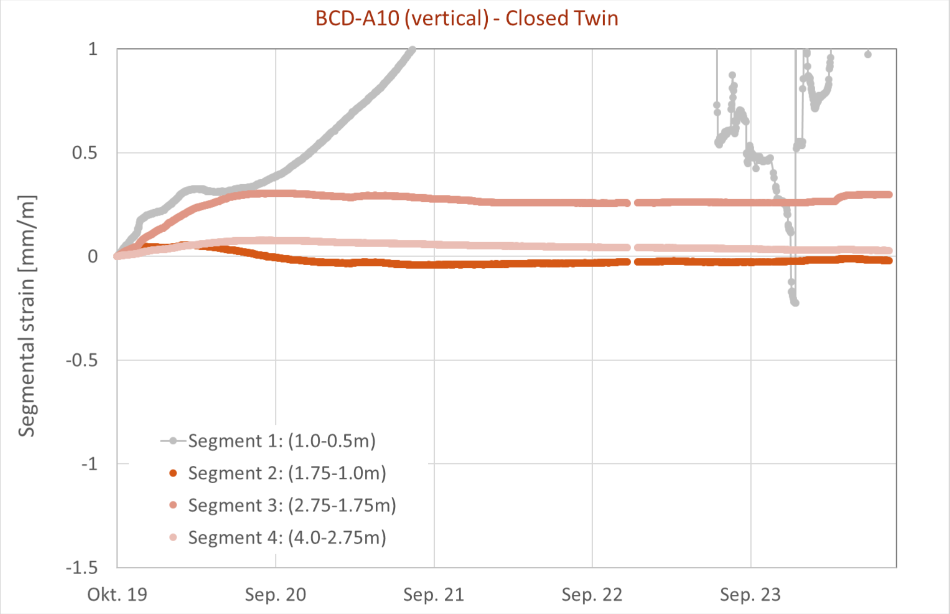


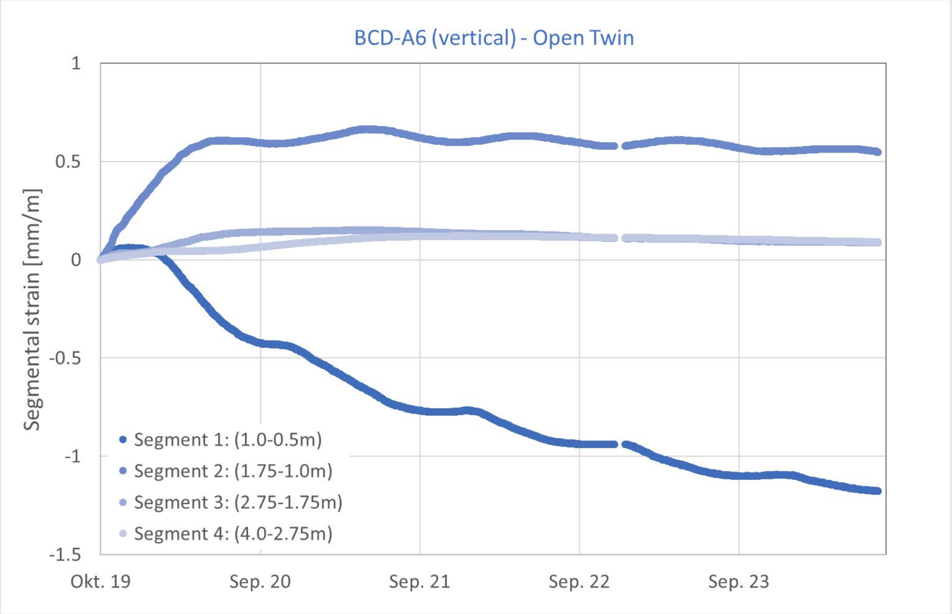
Ongoing compression of host rock within first meter

Distance > 175cm to the surface: no effect of climatic conditions

Segmental strains under the niches







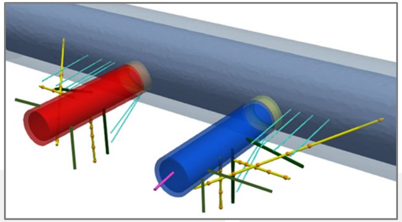
Strains remain low

Stationary conditions after ~ 1 year

Ongoing compression of host rock within first meter

Distance > 175cm to the surface: no effect of climatic conditions

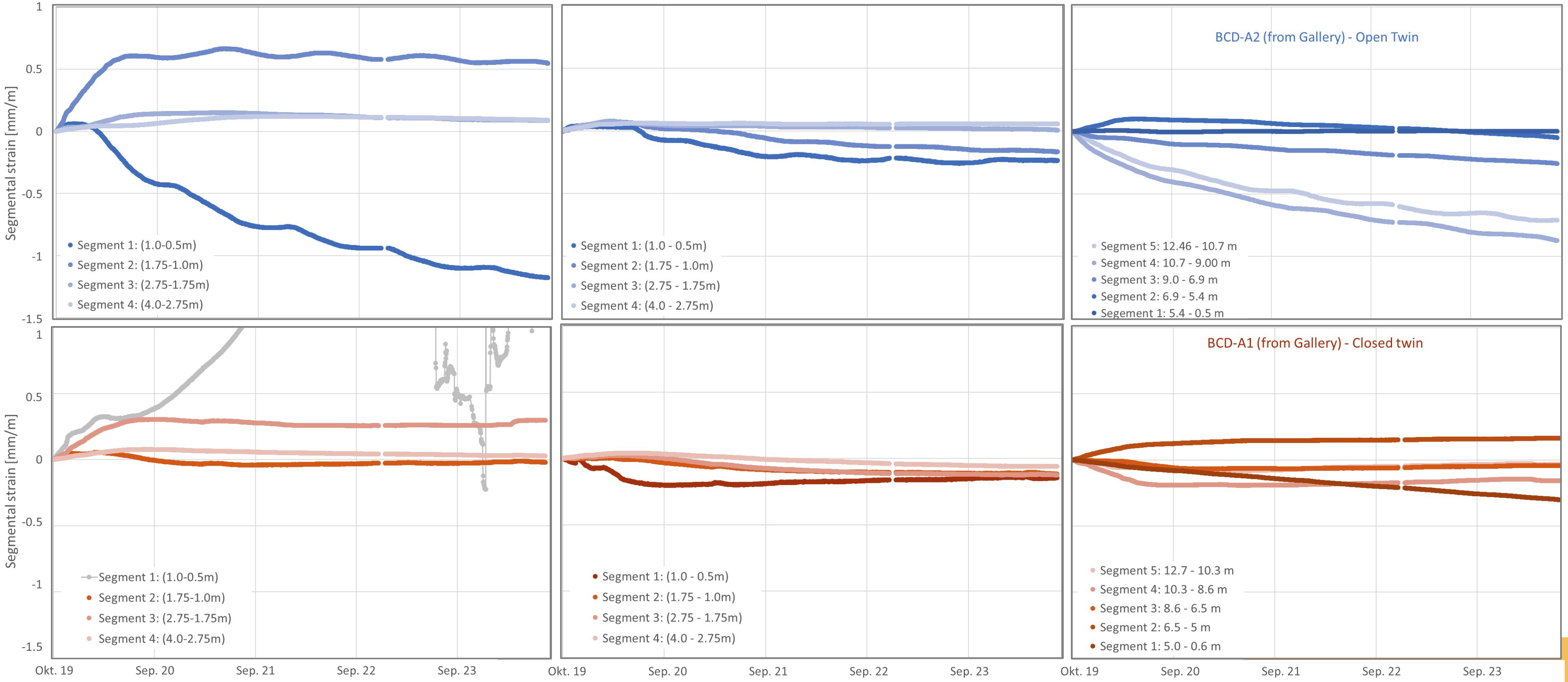
Segmental strains around the niches



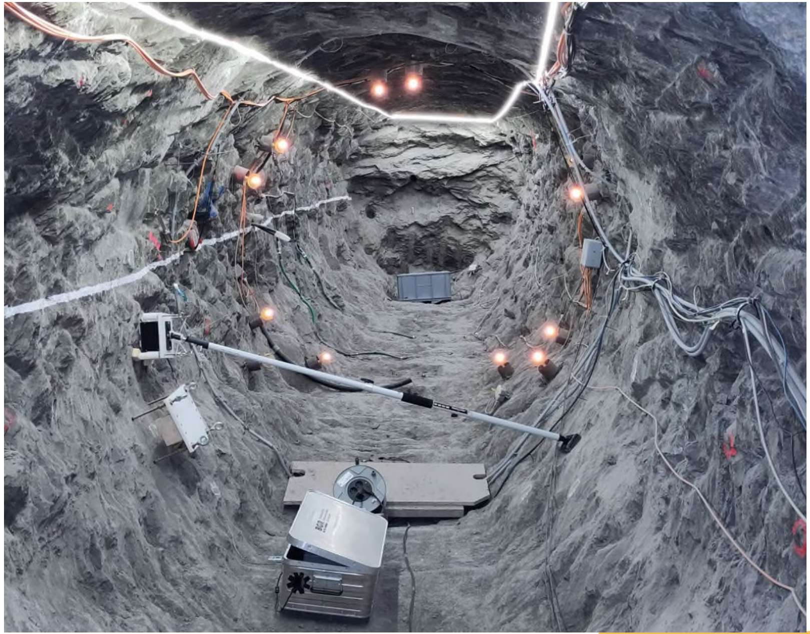
Vertical direction

Horizontal direction

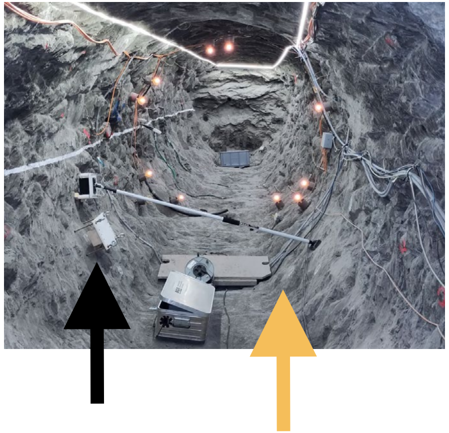
Drilled from gallery

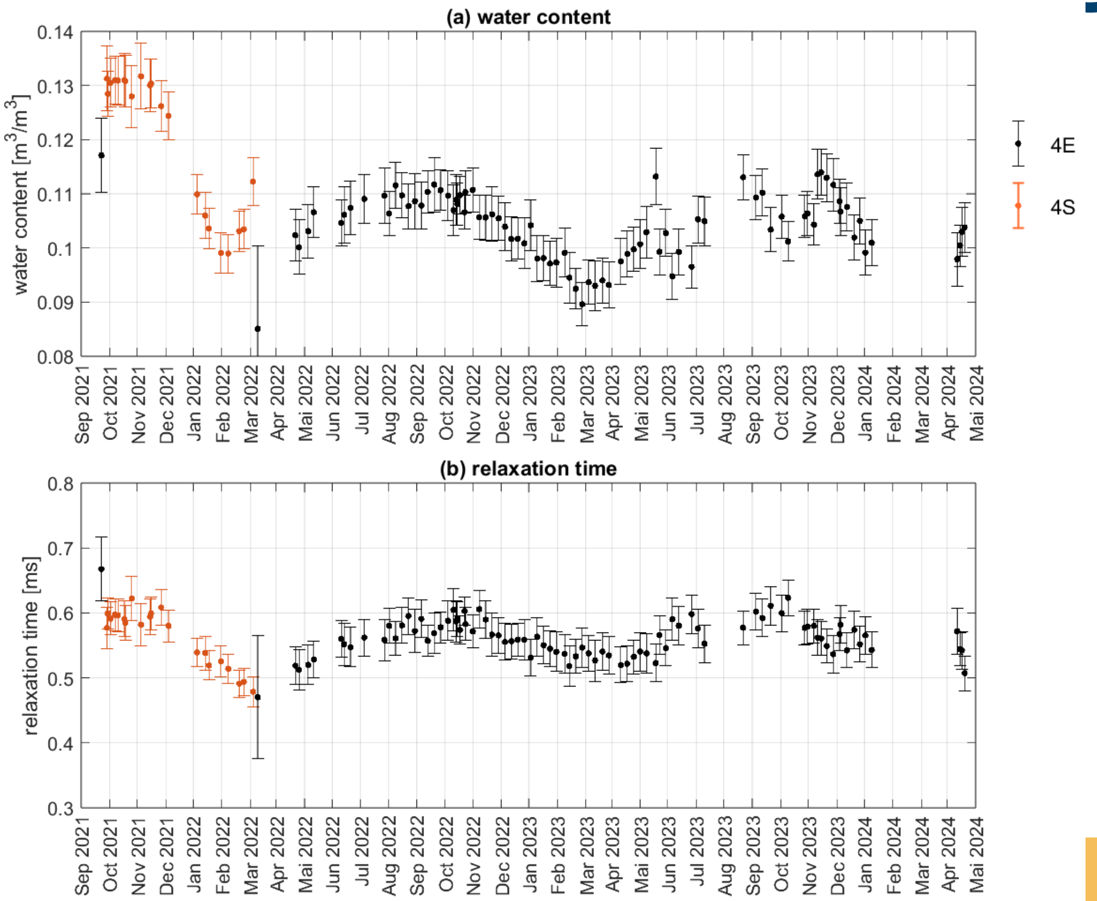


Single-Sided Nuclear Magnetic Resonance in the CD-A niches

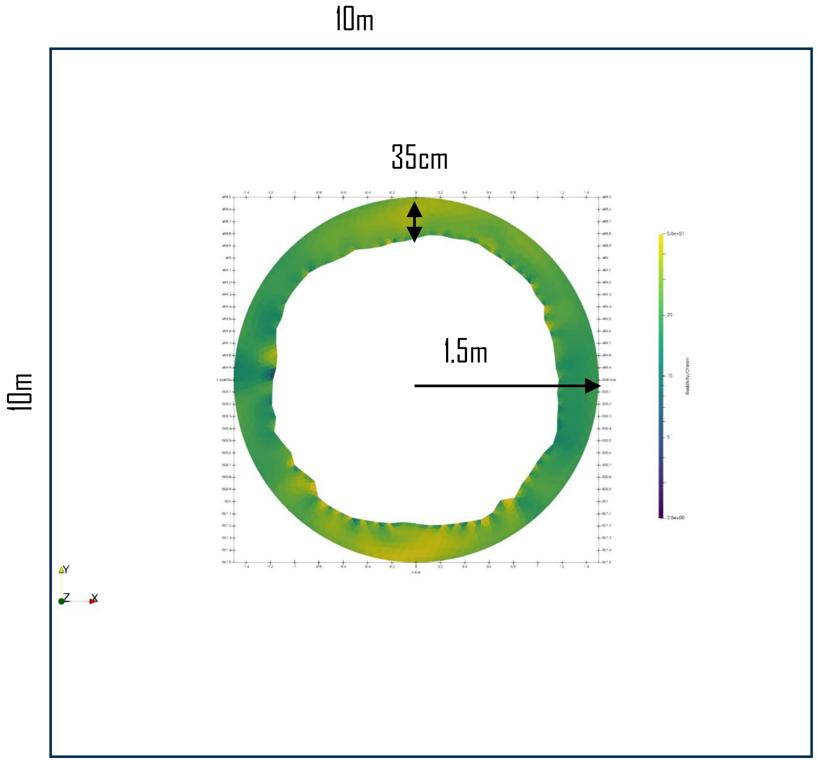


Weekly program on 4E in the open niche





Evaluation of ERT data



- Data on 10m by 10m
- Reliable range:

1.5m radius (green in the image),

35cm in the rock – only use this range!!

- Definition x/y (in model x/z)
- Mesh for ERT evaluation not equal to FEM mesh
- Data available daily (vtk)
- Data set includes several parameters (resistivity, delta resistivity, log resistivity, phase angle…)
- Data on exact circular shape

CD-A Experiment: Models

3D model:

- Modelling of excavation (gallery, niches)
- Modelling of long-term effects due to excavation and desaturation

2D (100m*100m):

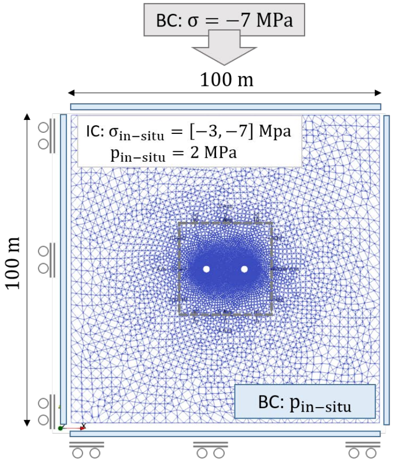
- Long-term effects due to desaturation
- Homogeneous material (EDZ, Host rock)

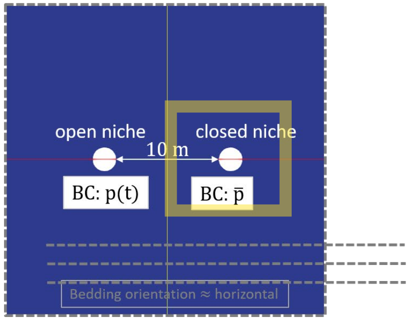
2D (10m*10m):

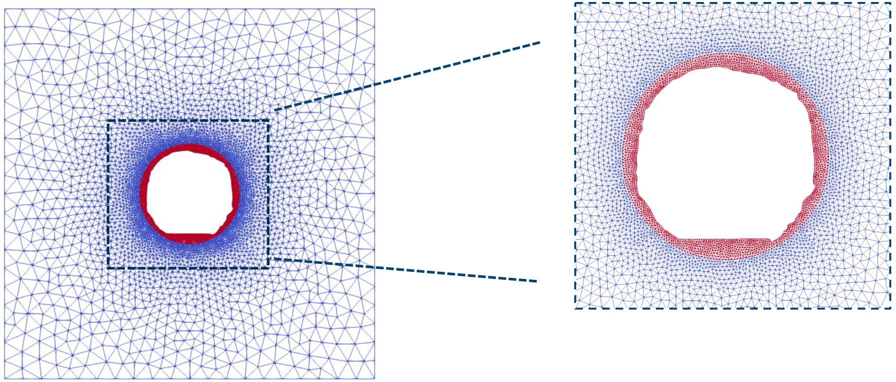
- **Long-term effects due to desaturation**
- **Use of ERT/NMR data**
- **Precise geometry**

2D Quarter:

- Ideal ring (DigBen)

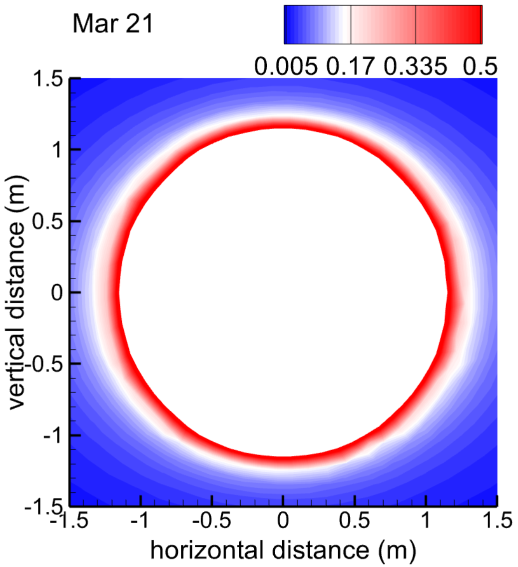




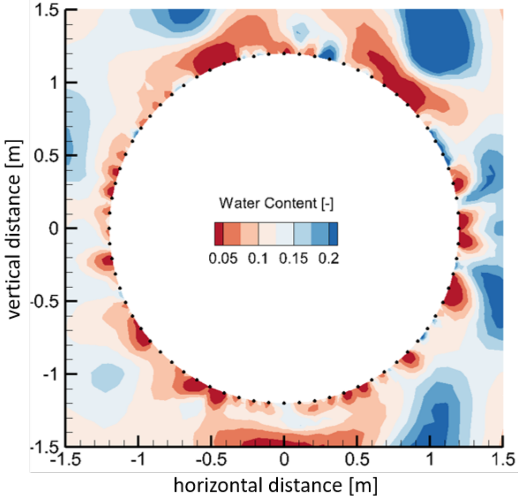


Combination of physical model and data set

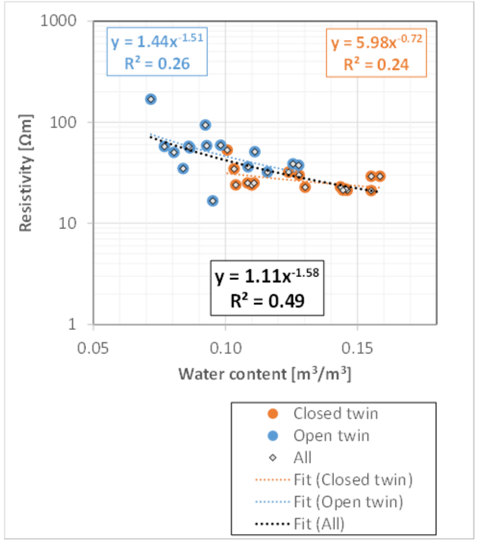
Numerical model:
saturation



ERT/NMR measurement:
Resistivity, water content



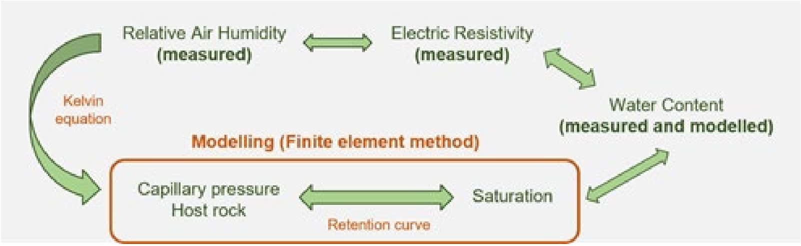
Relation between
Resistivity and water content



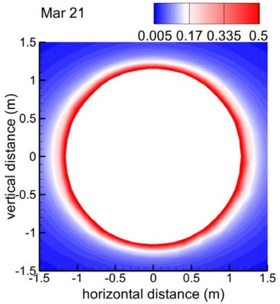
Combination of physical model and data set

- **Numerical (physical) model:**
  - Process variables (pressure, deformation, saturation)
  - Material parameters (permeability, porosity)

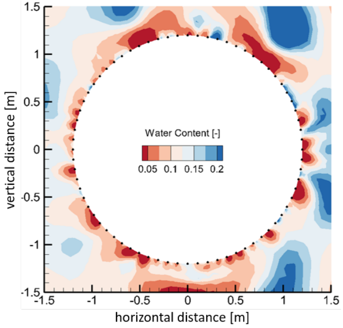
- **Data set:**
  - Electrical resistivity / Water content
  - Related to:
    - Saturation (process variable)
    - Permeability (material parameter)



- Adaptation of mesh during ML routine?



Numerical model: saturation



Data set: Eletrical resistivity / Water content

From our side, I think it would be good to start with the following:

- If you have any publications or notes on the model, it would help us get a better understanding.

https://rdcu.be/ea1vq

- If possible, could you provide access to a repository for your model in OpenGeoSys, ideally with instructions on how to set it up and get it running?

- Lastly, are you able to share the data, or is it subject to any licensing restrictions?

There’s no rush on any of this. As usual, the end of the year is quite busy for us, so we won’t be able to dive deeply into it in the coming months. However, it would be great to have a starting point ready for when we can find the time to explore further.